

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 12

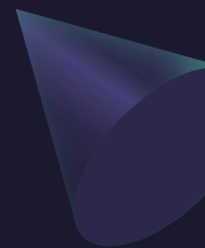


Any Reminders? Any Questions?

- I have regular office hours Mondays via Webex 1-2pm
- I have regular office hours Thursdays in-person 3:30-4:30pm
- Calc 3 Calc Night: MONT 104 at 6:30-8:30pm on Thursdays!



ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

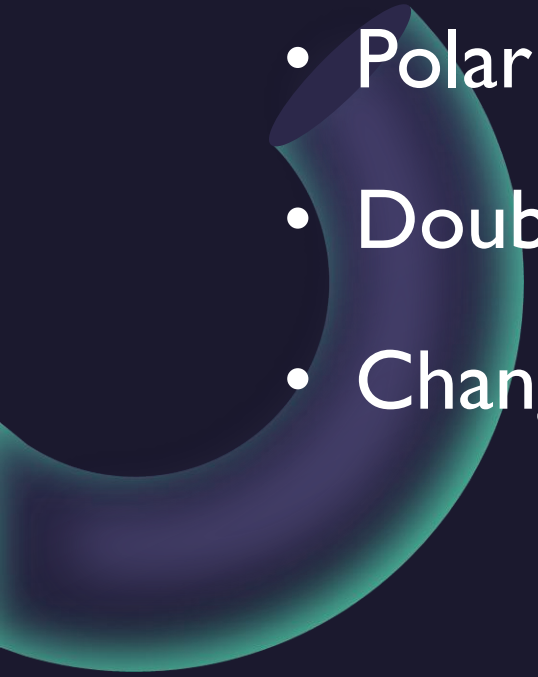
Instructor: Álvaro Lozano-Robledo

Double Integrals in Polar





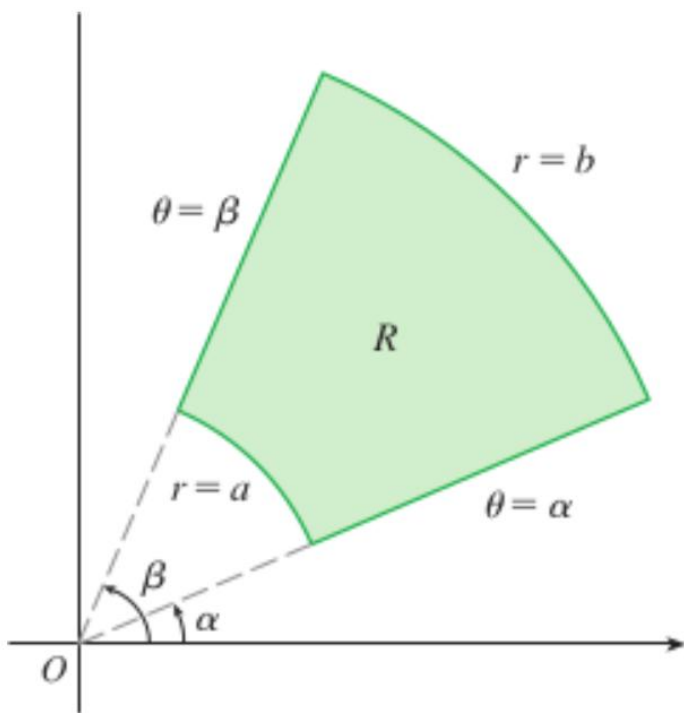
Today – Double Integrals in Polar Coordinates!

- 
- Polar Coordinates
 - Double Integrals in Polar Coordinates
 - Change from Cartesian to Polar

Double Integrals in Polar Coordinates

If f is continuous on a polar rectangle R given by $0 \leq a \leq r \leq b$, $\alpha \leq \theta \leq \beta$, where $0 \leq \beta - \alpha \leq 2\pi$, then

$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta$$



$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ r^2 = x^2 + y^2 \end{cases}$$

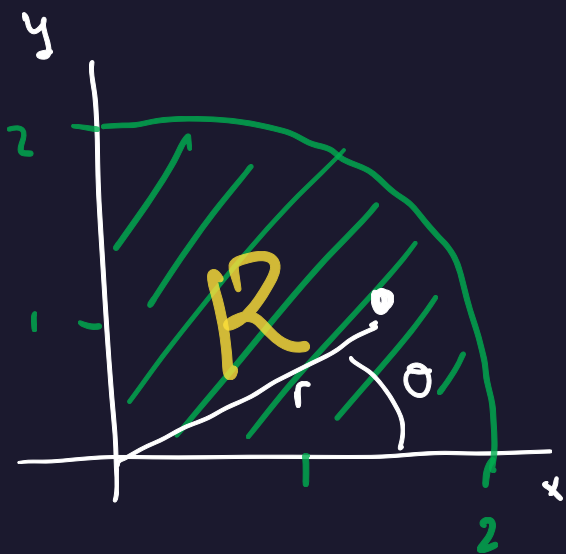
!!

Example: Evaluate the value of the double integral

$$I = \iint_R (x + x^2 + y^2) dA$$

where R is the first quadrant of a circle of radius 2.

$$R = \{(r, \theta) : 0 \leq r \leq 2, 0 \leq \theta \leq \pi/2\}$$



$$I = \int_0^{\pi/2} \int_0^2 (r \cos \theta + r^2) \cdot r \cdot dr d\theta$$

$$= \int_0^{\pi/2} \int_0^2 r^2 \cos \theta + r^3 dr d\theta$$

$$= \int_0^{\pi/2} \left[\frac{r^3}{3} \cos \theta + \frac{r^4}{4} \right]_0^2 d\theta = \int_0^{\pi/2} \left(\frac{8}{3} \cos \theta + 4 \right) d\theta$$

Example: Evaluate the value of the double integral

$$\iint_R (x + x^2 + y^2) dA$$

where R is the first quadrant of a circle of radius 2.

$$\int_0^{\pi/2} \left(\frac{8}{3} \cos \theta + 4 \right) d\theta$$

$$= \left(\frac{8}{3} \sin \theta + 4\theta \right) \bigg|_0^{\pi/2} = \frac{8}{3} \sin \frac{\pi}{2} + 4 \frac{\pi}{2} - \left(\frac{8}{3} \sin 0 + 4 \cdot 0 \right)$$

$$= \frac{8}{3} + 2\pi = \boxed{\frac{8}{3} + 2\pi}$$

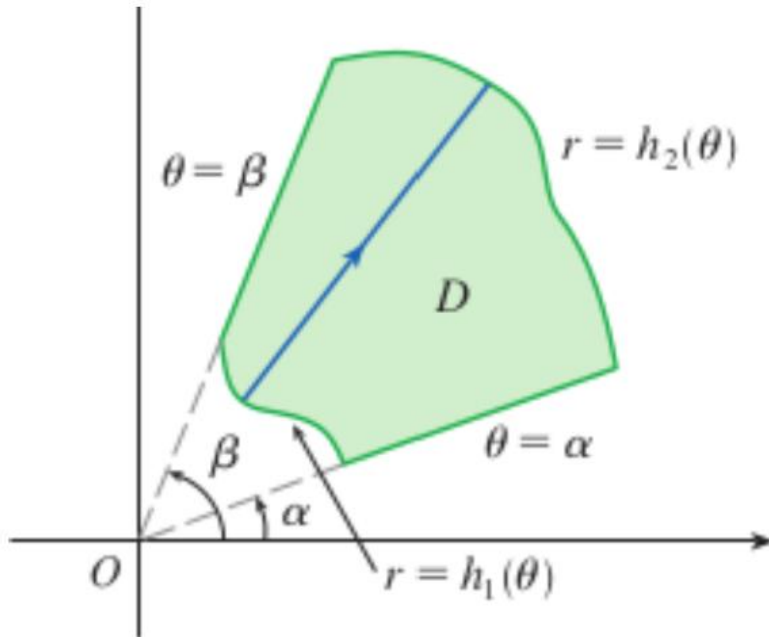
$$\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$$

Integrals in Polar Coordinates of Type I

$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$$

3 If f is continuous on a polar region of the form

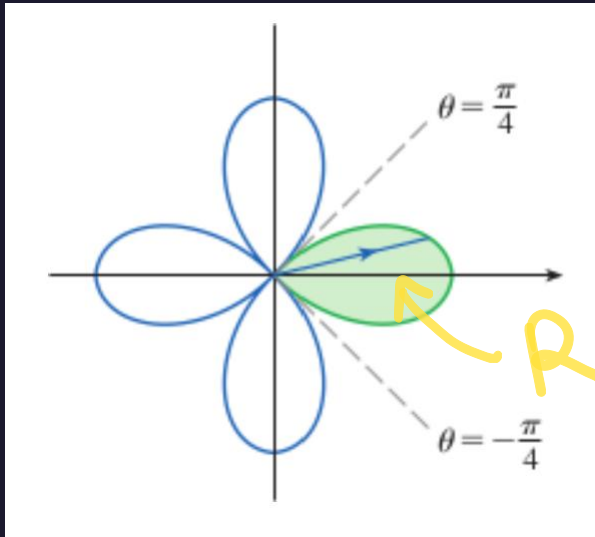
$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$$



$$\iint_D f(x, y) dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta$$

Example: Find the area enclosed by one loop of the four-leaved rose

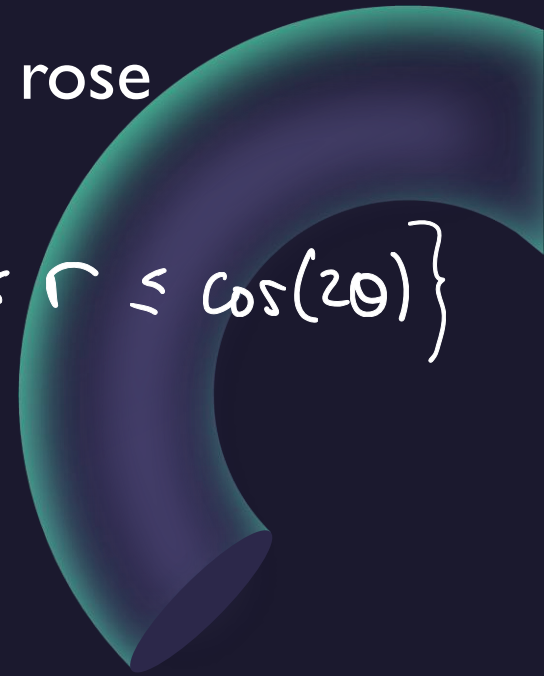
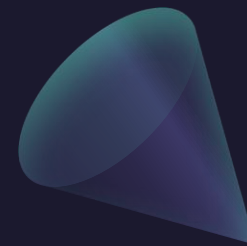
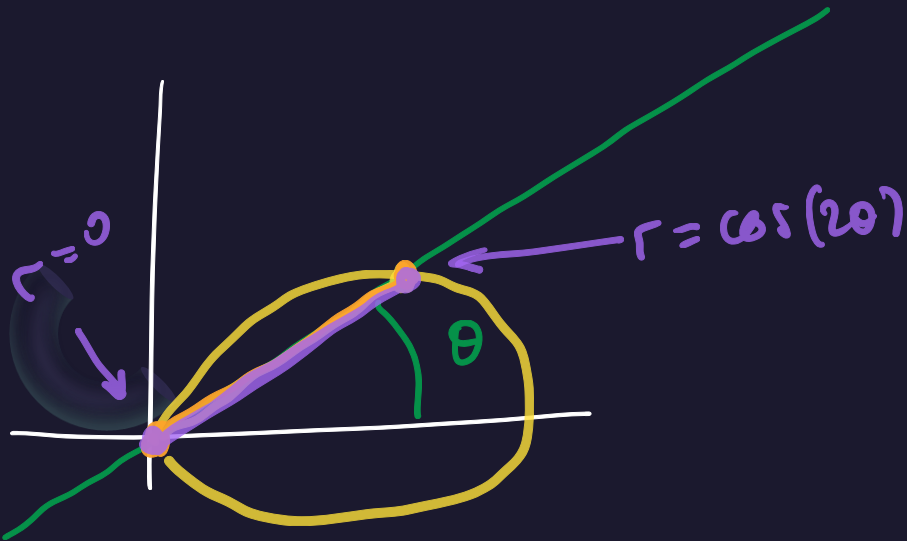
$$r = \cos(2\theta)$$



$$R = \left\{ (r, \theta) : -\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}, 0 \leq r \leq \cos(2\theta) \right\}$$

$$A(R) = \iint_R 1 \, dA$$

$$= \int_{-\pi/4}^{\pi/4} \int_0^{\cos(2\theta)} 1 \cdot r \cdot dr \, d\theta$$



Example: Find the area enclosed by one loop of the four-leaved rose

$$r = \cos(2\theta)$$

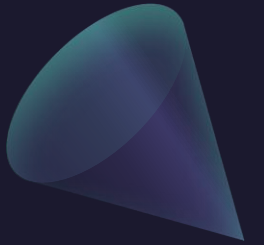
$$\int_{-\pi/4}^{\pi/4} \int_0^{\cos(2\theta)} r \, dr \, d\theta = \int_{-\pi/4}^{\pi/4} \left. \frac{r^2}{2} \right|_0^{\cos(2\theta)} d\theta$$

$$= \int_{-\pi/4}^{\pi/4} \frac{\cos^2(2\theta)}{2} d\theta = \int_{-\pi/4}^{\pi/4} \frac{1}{2} \cdot \frac{1 + \cos(4\theta)}{2} d\theta$$

$$= \frac{1}{4} \theta + \frac{\sin(4\theta)}{16} \Big|_{-\pi/4}^{\pi/4} = \boxed{\dots} \quad \frac{1}{4} \frac{\pi}{4} + \frac{1}{4} \frac{\pi}{4} = \boxed{\pi/8}$$

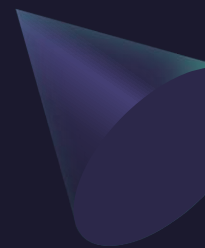
$\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}$

Questions?





ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Triple Integrals



TRIPLE

Today – ~~Double~~ Integrals in Polar Coordinates!

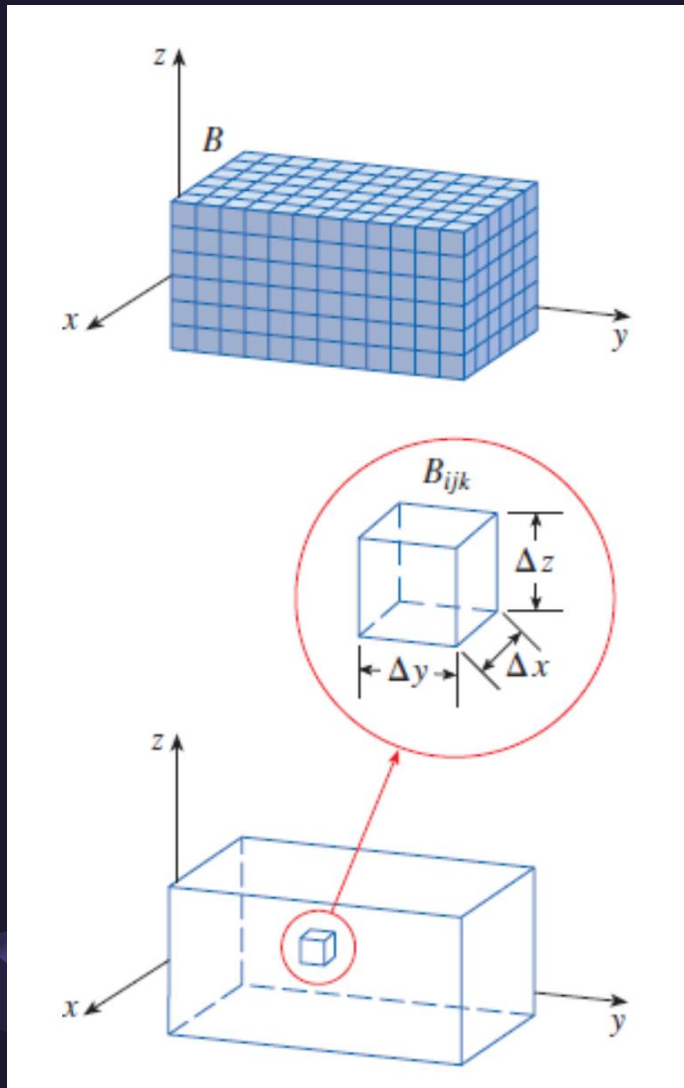
- Triple Integrals in Boxes
- Triple Integrals in Regions
- Changing the Order of Integration

$$f(x, y, z)$$



$$\iiint_R f(x, y, z) \, dV$$

Riemann Integral in 3 Variables



$$B = \{(x, y, z) \mid a \leq x \leq b, c \leq y \leq d, r \leq z \leq s\}$$

The **triple integral** of f over the box B is

$$\iiint_B f(x, y, z) \, dV = \lim_{l, m, n \rightarrow \infty} \sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \, \Delta V$$

if this limit exists.

Riemann Integral in 3 Variables

$$\{(x, y, z) \mid$$

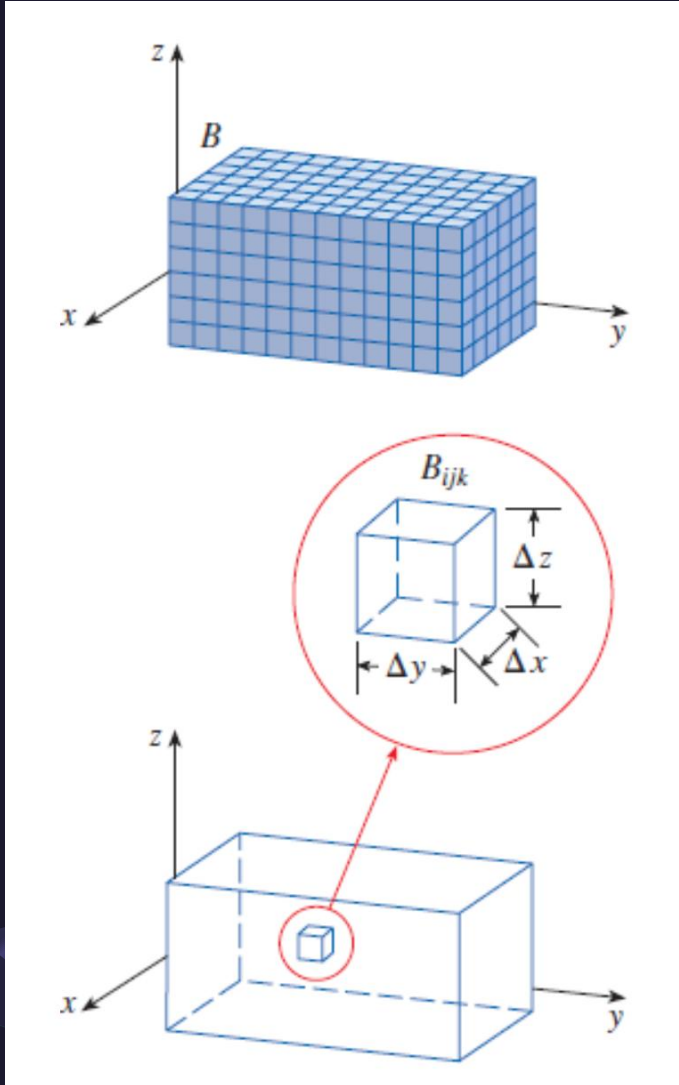
$$[a, b] \times [c, d] \times [r, s]$$

$$B = \{(x, y, z) \mid a \leq x \leq b, c \leq y \leq d, r \leq z \leq s\}$$

4 Fubini's Theorem for Triple Integrals

If f is continuous on the rectangular box $B = [a, b] \times [c, d] \times [r, s]$, then

$$\iiint_B f(x, y, z) \, dV = \int_r^s \int_c^d \int_a^b f(x, y, z) \, dx \, dy \, dz$$



Example: Evaluate the value of the triple integral

$$\iiint_B xyz^2 dV$$

where B is the box $[0,1] \times [-1,2] \times [0,3]$.

$$\int_0^3 \int_{-1}^2 \int_0^1 xyz^2 dx dy dz = \iint \frac{x^2}{2} y z^2 \Big|_0^1 dy dz$$

$$= \int_0^3 \int_{-1}^2 \frac{1}{2} y z^2 dy dz = \int_0^3 \frac{1}{4} y^2 z^2 \Big|_{-1}^2 dz$$

$$= \int_0^3 \frac{1}{4} z^2 z^2 - \frac{(-1)^2}{4} z^2 dz = \int_0^3 \frac{3}{4} z^2 dz = \frac{z^3}{4} \Big|_0^3 = \boxed{\frac{27}{4}}$$

Example: Evaluate the value of the triple integral

$$\iiint_B xyz^2 dV$$

where B is the box $[0,1] \times [-1,2] \times [0,3]$.

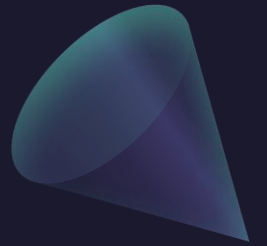
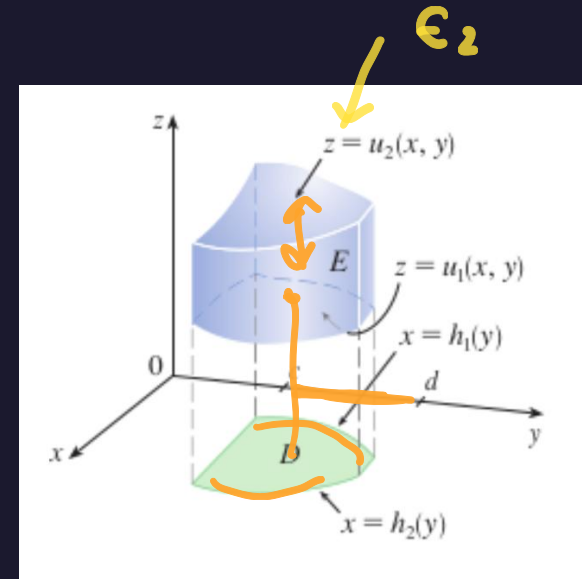
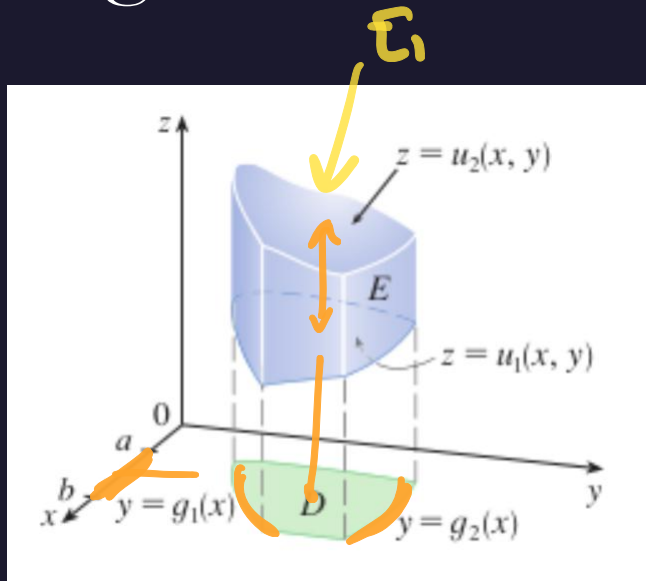
$$\text{Volume} = 1 \cdot 3 \cdot 3 = 9$$

$$xyz^2 = x \cdot y \cdot z^2$$

$$\iiint_B 1 dV = V(B) = 9$$

$$\int_0^3 \int_{-1}^2 \int_0^1 xyz^2 dx dy dz = \left(\int_0^3 z^2 dz \right) \cdot \left(\int_{-1}^2 y dy \right) \cdot \left(\int_0^1 x dx \right)$$

Triple Integrals in General Regions – Define the Region



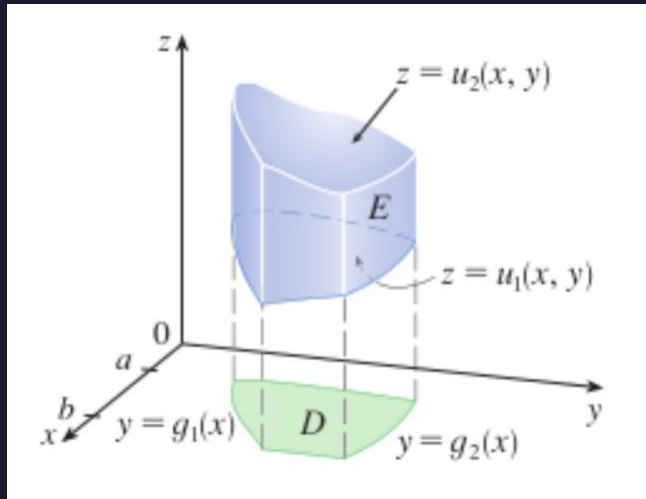
$$E_1 = \{(x, y, z) : \underbrace{a \leq x \leq b, g_1(x) \leq y \leq g_2(x)}_{\text{Type I}}, u_1(x, y) \leq z \leq u_2(x, y)\}$$

$$E_2 = \{(x, y, z) : \underbrace{c \leq y \leq d, h_1(y) \leq x \leq h_2(y)}_{\text{Type II}}, u_1(x, y) \leq z \leq u_2(x, y)\}$$

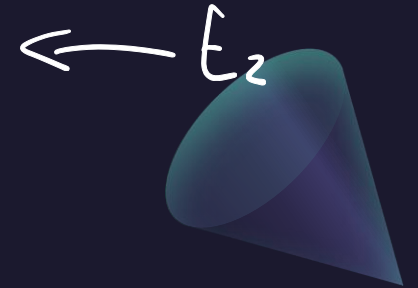
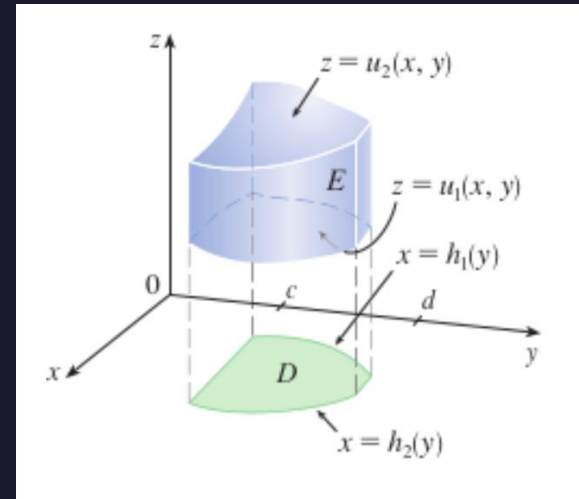
TYPE II

Triple Integrals in General Regions – Set up the Integral

E_1 →



$f(x, y, z)$

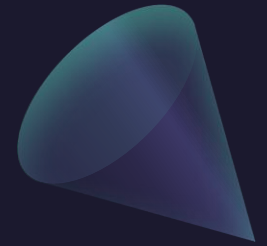
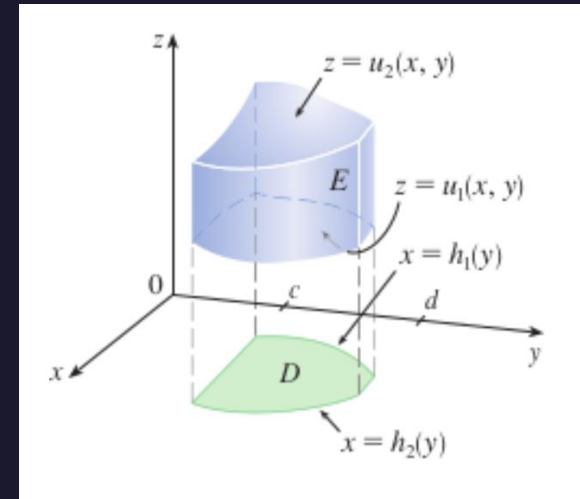
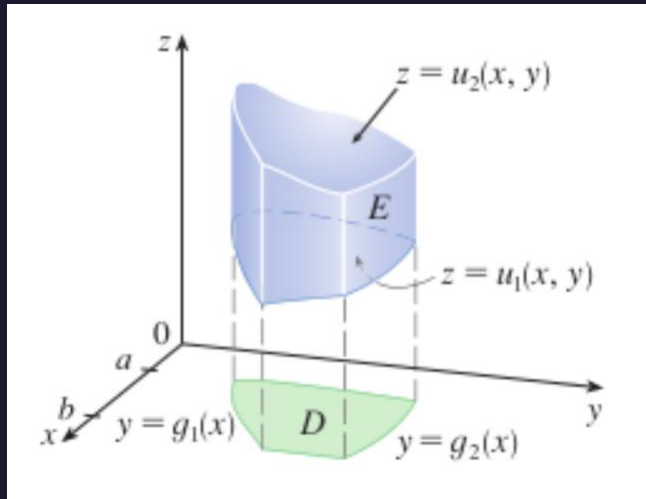


On E_1 ,
$$\int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z) \, dz \, dy \, dx$$

On E_2 ,
$$\int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z) \, dz \, dx \, dy$$

Type 1
Triple

Triple Integrals in General Regions



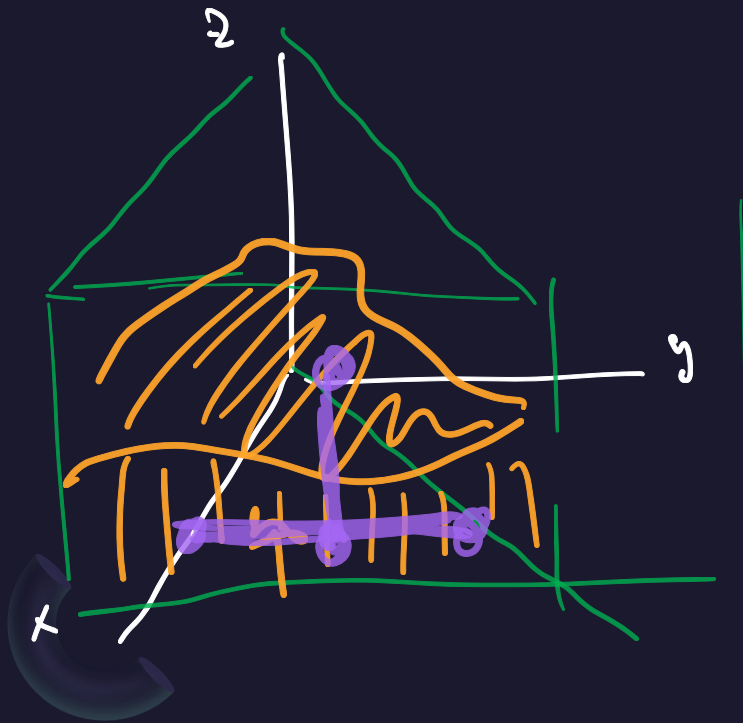
$$\iiint_E f(x, y, z) dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz dy dx$$

$$\iiint_E f(x, y, z) dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz dx dy$$

Example: Evaluate the value of the triple integral

$$I = \iiint_E z \, dV$$

where E is the solid in the first octant bounded by the surface $z = 12xy$ and the planes $y = x$ and $x = 1$.



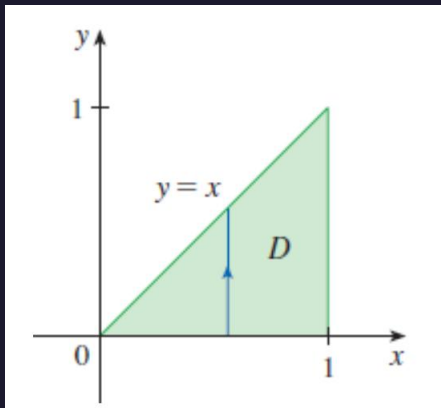
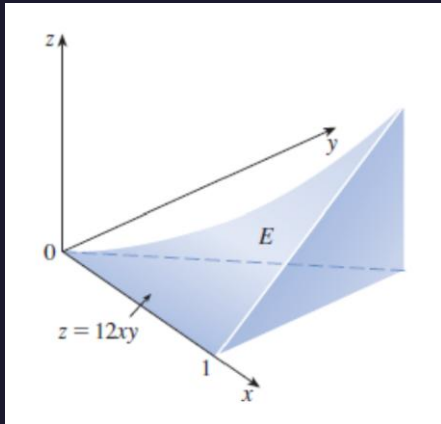
$$E = \left\{ (x, y, z) : \begin{array}{l} 0 \leq x \leq 1, \\ 0 \leq y \leq x, \\ 0 \leq z \leq 12xy \end{array} \right\}$$

$$I = \int_0^1 \int_0^x \int_0^{12xy} z \cdot dz \, dy \, dx$$

Example: Evaluate the value of the triple integral

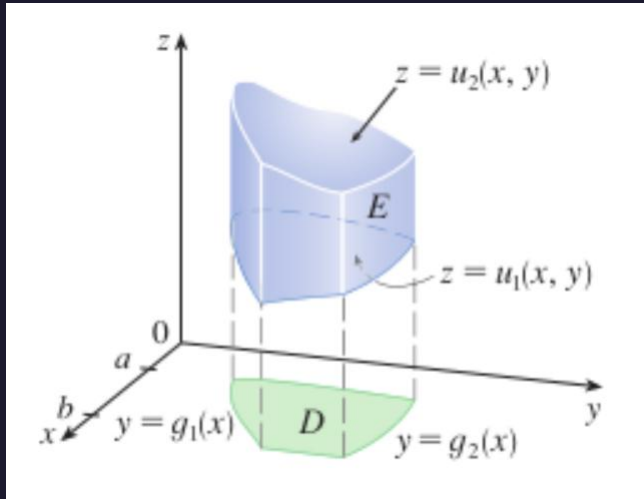
$$\iiint_E z \, dV$$

where E is the solid in the first octant bounded by the surface $z = 12xy$ and the planes $y = x$ and $x = 1$.

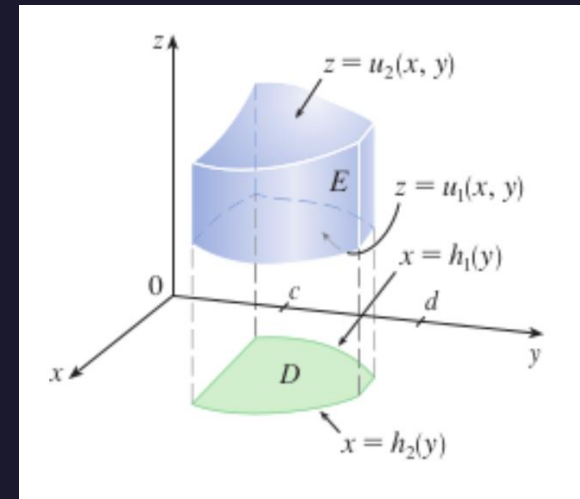


$$\begin{aligned} & \int_0^1 \int_0^x \int_0^{12xy} z \, dz \, dy \, dx \\ &= \int_0^1 \int_0^x \left. \frac{z^2}{2} \right|_0^{12xy} dy \, dx = \int_0^1 \int_0^x \frac{12^2 x^2 y^2}{2} dy \, dx \\ &= \int_0^1 \left. \frac{12^2}{2} x^2 \frac{y^3}{3} \right|_0^x dx = \int_0^1 \frac{12^2}{6} \frac{x^6}{6} dx = \dots \end{aligned}$$

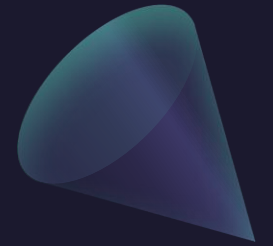
Triple Integrals in General Regions: Type I, ...



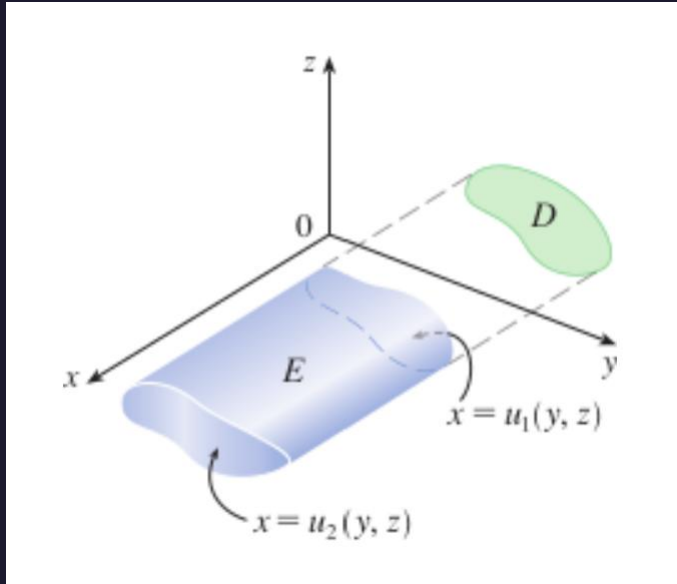
$$\iiint_E f(x, y, z) \, dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \, dx \, dy$$



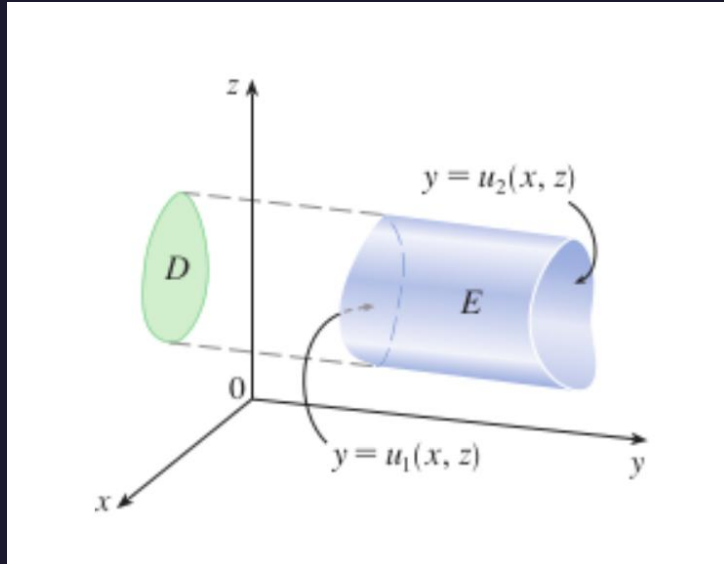
$$\iiint_E f(x, y, z) \, dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \, dx \, dy$$



Triple Integrals in General Regions: Type II



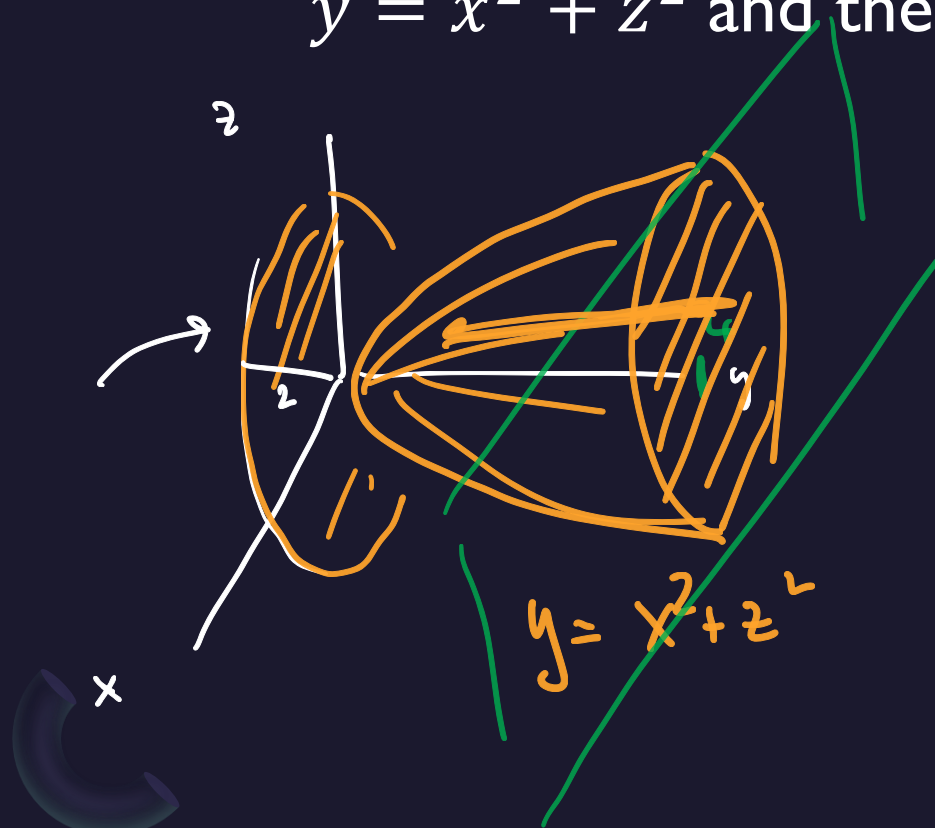
Triple Integrals in General Regions: Type III



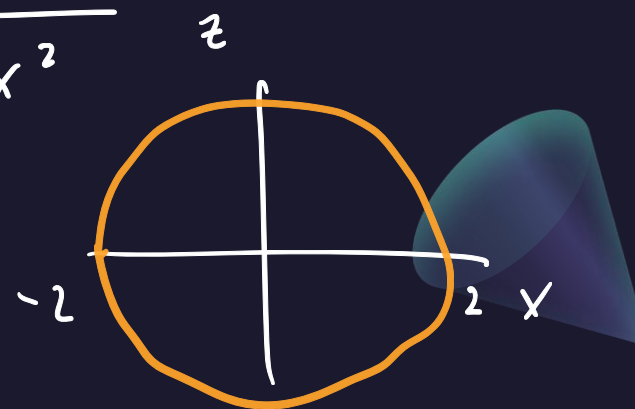
Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + y^2} \, dV$$

where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.



$$E = \left\{ (x, y, z) : \begin{array}{l} -2 \leq x \leq 2 \\ -\sqrt{4-x^2} \leq z \leq \sqrt{4-x^2} \\ x^2 + z^2 \leq y \leq 4 \end{array} \right\}$$
$$x^2 + z^2 = 4$$
$$z = \pm \sqrt{4-x^2}$$



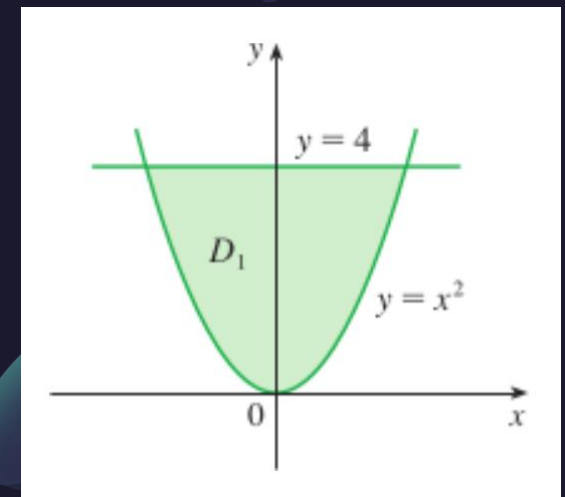
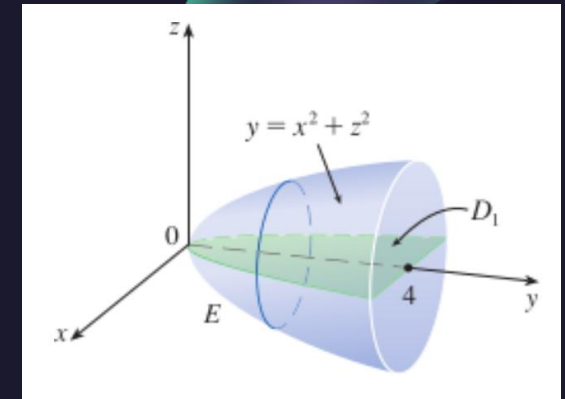
Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + y^2} \, dV$$

where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{x^2+z^2}^4 \sqrt{x^2+y^2} \, dy \, dz \, dx$$

WAIT! CYLINDRICAL
COORDS!



Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

Thank you

Until next time.

